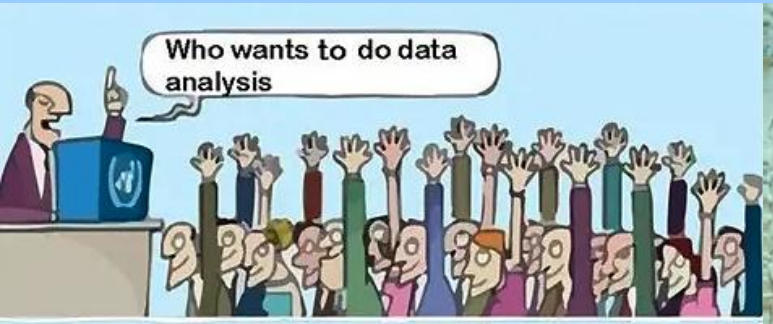
香港教育大學
The Education University of Hong Kong

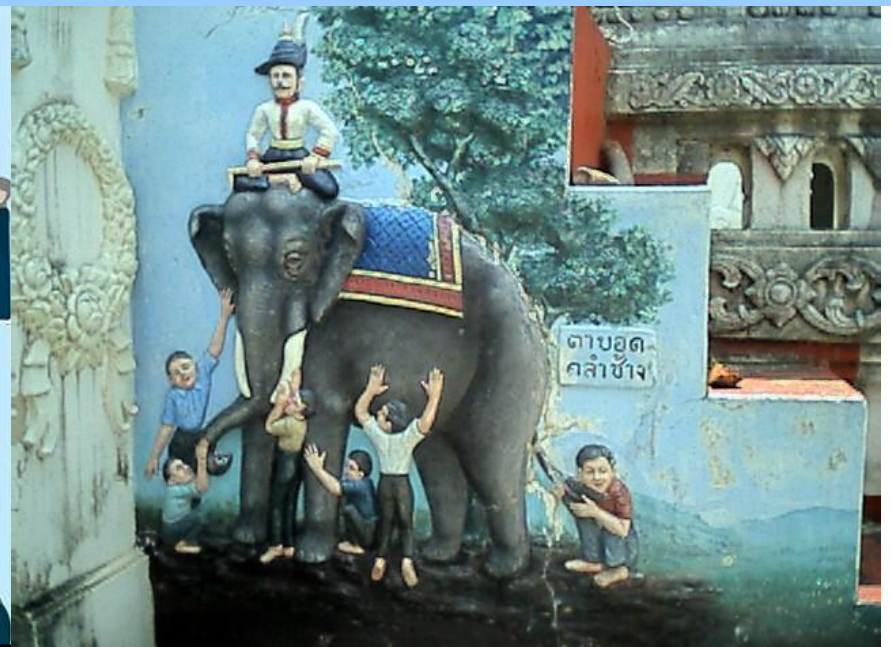
MSc(AIET) 2025-26

MTH6184 Data Mining in STEM Education

Chapter 2. Data Preprocessing

Instructor: Philip Yu





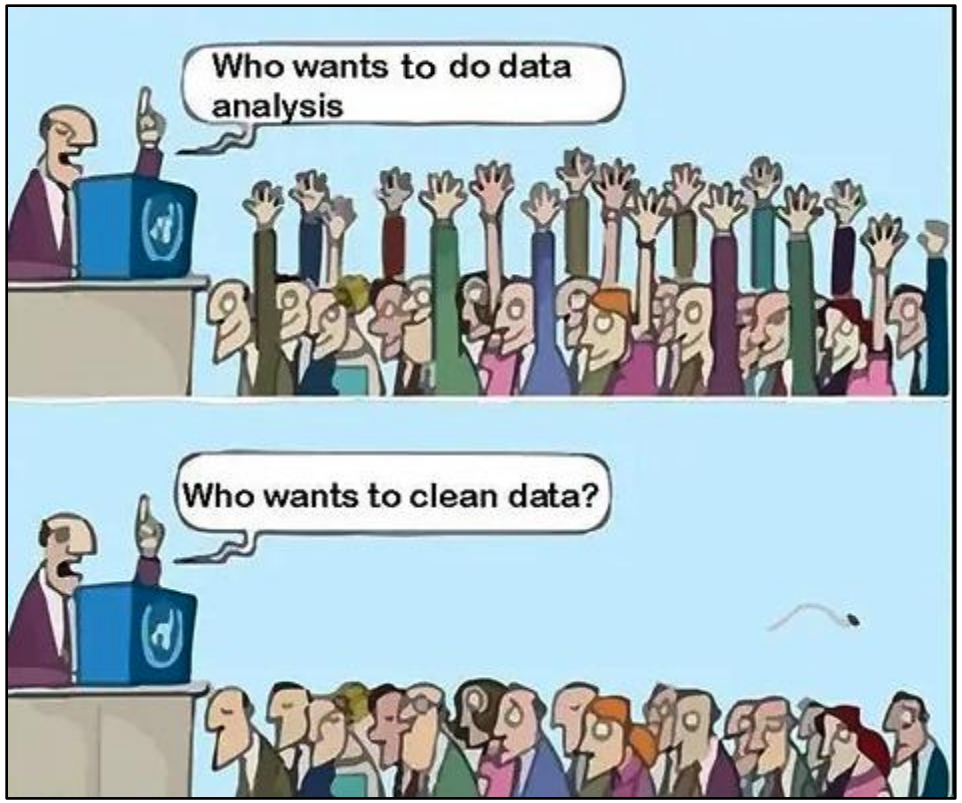
"Success depends upon previous preparation, and without such preparation there is sure to be failure."—Confucius

凡事豫（預）則立，不豫（預）則廢。
《禮記，中庸》

Chapter Outcomes

After completing this chapter, you should:

- Understand different types of data
- Know major tasks in data preprocessing including
  - Data cleaning (& Data exploration)
  - Feature transformation
  - Feature selection (or Dimension reduction)
- Be aware of ethical issues



Types of Data

Structured:

- Multidimensional data
- Transactional data
- Sequential data
- Spatial-temporal data

Unstructured:

- Multimedia data
- Textual data
- Network data

Data Format

Data Source	Structured	Unstructured
Internal	**Human-Generated** • Survey ratings • Aptitude testing **Machine-Generated** • Web metrics from Web logs • Product purchase from sales Records • Process control measures	**Human-Generated** • Emails, letters, text messages • Audio transcripts • Customer comments • Voicemails • Corporate video/communications • Pictures, illustrations • Employee reviews
External	**Human-Generated** • Number of Retweets, Facebook likes, Google Plus +1s • Ratings on Yelp • Patient ratings ratings **Machine-Generated** • GPS for tweets • Time of tweet/updates/postings	**Human-Generated** • Content of social media updates • Comments in online forums • Comments on Yelp • Video reviews • Pinterest images • Surveillance video

https://businessoverbroadway.com/wp-content/uploads/2014/07/Data-Definition-Framework.png

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Multidimensional Data

- Collection of data objects and their attributes
- An attribute is a characteristic of an object
  - Examples: eye color of a person, temperature, etc.
- Attribute is also known as variable or feature
- Object is also known as record, point, case, sample, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Four Types of Attributes (Variables)

Attribute Type	Description	Examples	Methods
Nominal ($= \neq$)	Use a label or name to distinguish one object from another	Staff ID, zip codes, eye color, sex: {*male, female*}	mode, contingency correlation, χ^2 test
Ordinal ($= \neq > <$)	Use values to provide the ordering of objects	hardness of minerals, {*good, better, best*}, grades, ranking scale 1-5	median, percentiles, rank correlation, runs tests, sign tests
Interval ($= \neq > < + -$)	Use units of measurement, but the origin is arbitrary	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, *t* and *F* tests
Ratio ($= \neq > < + - * /$)	Use units of measurement, but the origin is not arbitrary	temperature in Kelvin, monetary quantities, age, counts, mass, length	geometric mean, harmonic mean, percent variation

- Which type does a binary variable belong to?

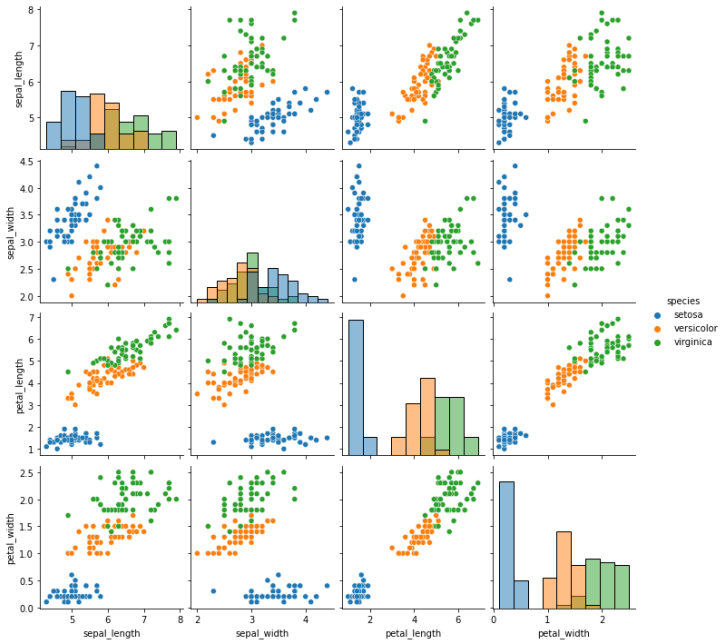
Data Visualization

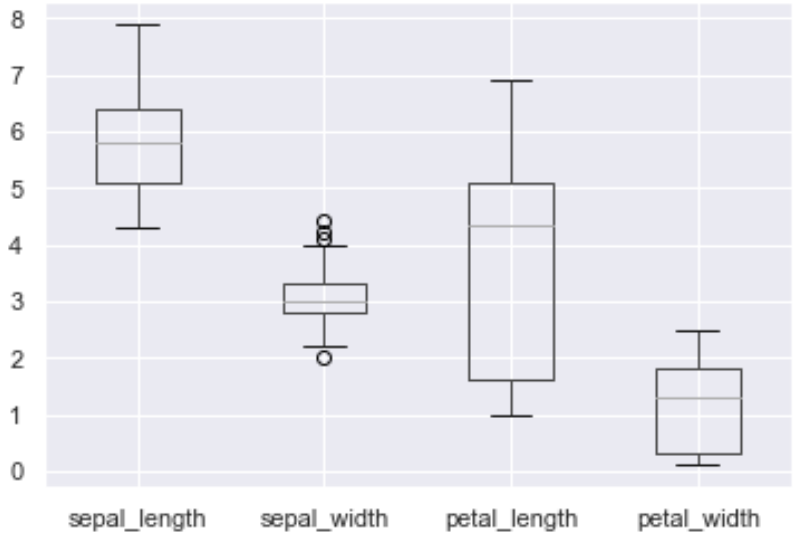
Construct appropriate graphs to examine patterns of data:

- Distribution:
  - Nominal and Ordinal: Bar charts, Pie charts
  - Interval & Ratio: Box plots; Histograms
- Trend:
  - Line charts
- Dependence:
  - Scatterplots (2D plots)
  - 3D plots

```
     Q1-1.5IQR   Q1   median  Q3   Q3+1.5IQR
                  |-----:-----|
  o      |--------|     :     |--------|    o  o
                  |-----:-----|
flier             <----------->           fliers
                       IQR
```

https://matplotlib.org/3.5.3/api/_as_gen/matplotlib.pyplot.boxplot.html





Envisage the Youth Development Blueprint:

Survey Summary and Advocacy

The Home & Youth Affairs Bureau will publish the first edition of the Youth Development Blueprint by 2022 year-end to outline the principles, objectives and actions of the Government in pursuing youth development. The Hong Kong Federation of Youth Groups' (HKFYG) Youth Research Centre surveyed 630 respondents aged 15 to 35 years old in October 2022 on their views. YHK shares the following findings.

- 630 respondents aged 15-35 years old were recently interviewed about their views about future development by HKFYG's Youth Research Centre, in tandem with the Youth Development Blueprint to be released by the Home and Youth Affairs Bureau (HYAB).
- Apart from HYAB's seven policy aspects, we suggest that equal opportunities and sustainable development are crucial areas to be included in the blueprint.
- Diversifying Hong Kong's economy is essential to youth development, such as providing more support to the newly emerged industries like the creative craftsmanship industry or sports industry. For talent development, response to the trend of "slashie" culture should take priority.
- Our survey reaffirms that the housing problem is of utmost

- 民政及青年事務局（HYAB）將於今年內公布首份《青年發展藍圖》。青協青年研究中心以問卷調查 630 名 15 至 35 歲的青年，以了解他們對《藍圖》及未來發展的看法。
- 除了局方提出的七大範疇，我們建議《藍圖》應同時涵蓋「平等機會」及「可持續發展」，以更全面反映香港青年未來發展的重要元素。
- 香港推動經濟多元化對青年發展至關重要，例如為創意工藝或體育產業等新興行業提供更多支持。為了培養更多人才，社會應接納及正面回應近年甚受年輕一代歡迎的「斜槓族」文化。
- 我們的調查再次發現，年輕人最關心住屋問題，建議增加青年宿舍的供應，設立青年租

- The average satisfaction score for Hong Kong's current situation for young people is 4.32 (on a 0-10 scale), which is below average.

improve employment and entrepreneurship opportunities by promoting upward mobility of young people (59.5%)

58.3%

- Respond to the home-buying opportunities and housing needs of young people (58.3%).

- Only 11.9% felt Hong Kong has enough opportunities for career development
- Only 12.9% said they are satisfied with their opportunities in Hong Kong

The top three issues that youth consider the most important are:

- Housing needs (53.3%)

53.3%

- Income (49.5%)

49.5%

- Emotional/mental health (30.8%)

30.8%

Regarding the importance of different youth policies, more than 40% of the respondents consider the following as very important:

- Promoting youth mental health (46.3%)

46.3%

- retaining talent (44.0%)

44%

- improving the housing conditions of young people (43.5%)

43.5%

The top three values that youth consider important are:

Empathy 42.9%

Respect 45.7%

Responsibility 48.7%

Our Advocacy

1. Don't miss equal opportunities and sustainable development

Apart from the HYAB's seven policy foci, we suggest that equal opportunities and sustainable development are crucial areas to be included in the Blueprint. Since more attention and from different ethnic qual treatment in the government's ethnic minority addressed. For r to the 17 SDGs of t youth development mutually supportive ould formulate an of youth in pment of Hong Kong tion, fostering youth- youth to participate

2. Foster diversified industry and talent development

Over the past years, Hong Kong has relied on the development of the four pillar industries, namely: financial services; trading and logistics; tourism, and producer and professional services. Diversifying Hong Kong's economy is essential to youth development. The government should provide more support to the newly emerged industries such ascreative craftsmanship or sports industries. For talent development, the trend of "slashie" culture should take priority. The government should improve the qualifications framework so that flexi-work experience can be accumulated and recognized.

3. Address the housing needs

Our survey reaffirms that the housing issue is of utmost concern among Hong Kong youth. The HKSARG needs to show their determination in creating housing ladder for young people by providing more affordable housing options. Our suggestions include, increasing the supply of youth hostels, and setting up Cash Allowance Trial Scheme to rent a flat.

4. Broaden Young People's Horizons

In order to let Hong Kong youth acquire a better understanding of the development of our country and the world, and develop a proper and holistic outlook, engaging more youth to participate in Mainland and international internship and exchange programmes should be the continuous work of the government. This will help strengthen the competitiveness of young people in Hong Kong. ■

For enquiries, pls contact Miss Chan Shui Ching / Unit-In-Charge at 3755 7040

Class Activity 1:
Data Exploration and Visualization

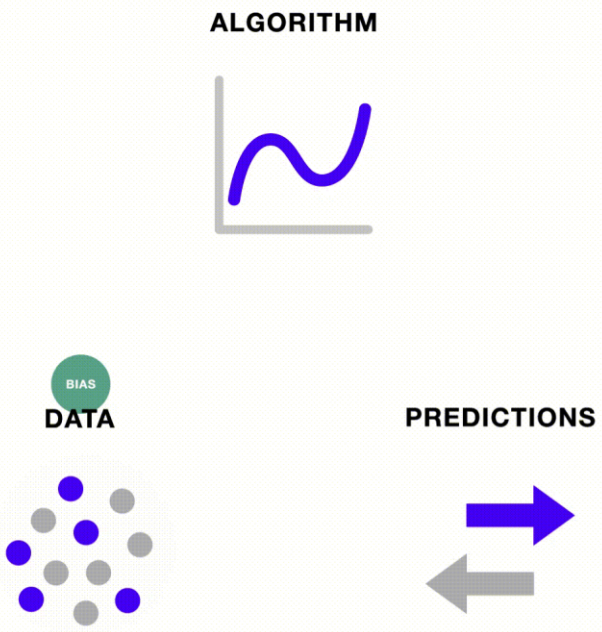
Why Data Preprocessing?

- Data in the real world is dirty!
  - Incomplete: missing values, lacking certain variables of interest, or containing only aggregate data
  - Noisy: containing errors, outliers or inaccurate values
  - Inconsistent: containing discrepancies in codes or names or out-of-range values
- Large and massive datasets
  - May contain irrelevant variables or old data.
- No quality data, no quality analytics
  - Quality decisions must be based on quality data

df.fillna(0)

	col1	col2	col3	col4	col5
1	2	5.0	3.0	6	?
2	9	?	9.0	0	7.0
3	19	17.0	?	9	?

	col1	col2	col3	col4	col5
1	2	5.0	3.0	6	0.0
2	9	0.0	9.0	0	7.0
3	19	17.0	0.0	9	0.0



Garbage In, Garbage Out

Major Steps of Data Preprocessing

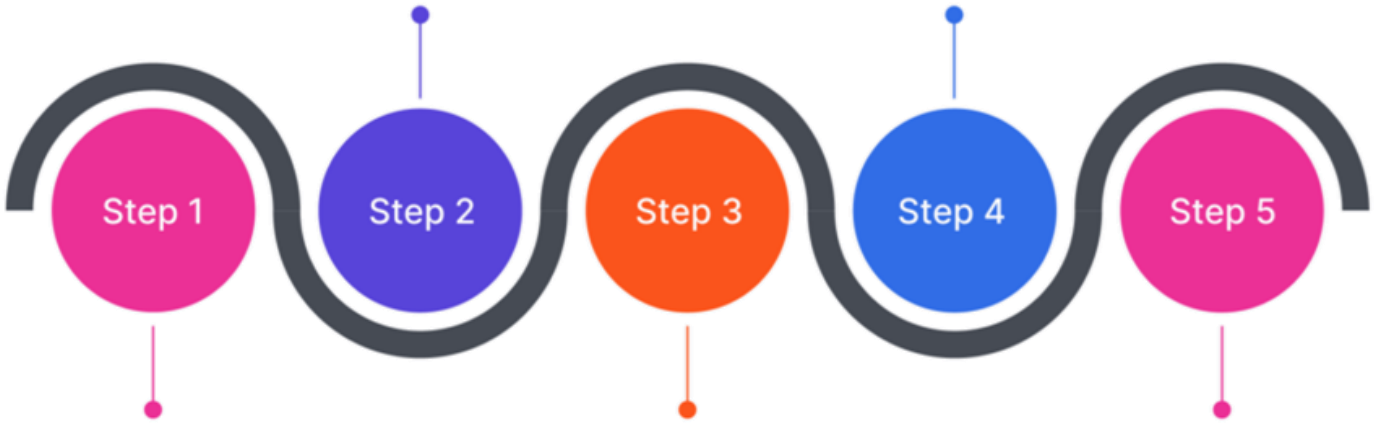
Data Cleaning

- Impute missing values
- Reduce noise in data
- Eliminate inconsistencies

Data Reduction

- Reduce number of variables
- Reduce number of cases
- Balance skewed data

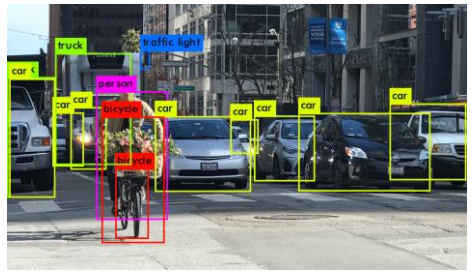
Real-world Data



Well-formed Data

Data Collection

- Collect data, Web scraping
- Select data, Annotating data
- Integrate data



Data Transformation

- Discretize / Aggregate data
- Extract / Transform features
- Select features

Data Splitting

- Training data
- Validation data
- Testing data

Data Cleaning

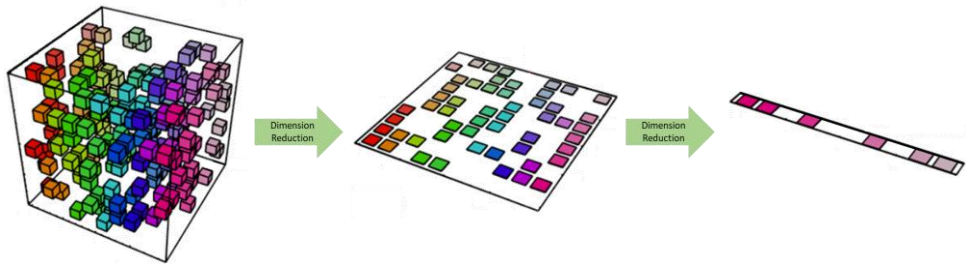
Possible kind of data problems:

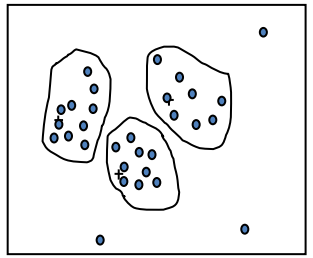
- Incomplete data (Missing values)
- Noisy data (Errors or Outliers)
  - Age = 120
- Data Inconsistencies (Coding or Naming)
  - Name = 'Philip Yu' for record 1; and

    Name = 'Philip Yeung' for record 2
  - Variable names, CUST_ID and CUSTOMER_ID
- Duplicated records
  - Job vacancies from two sources: JobsDB and LinkedIn;
    Patient records from hospitals

How to Detect Data Problems?

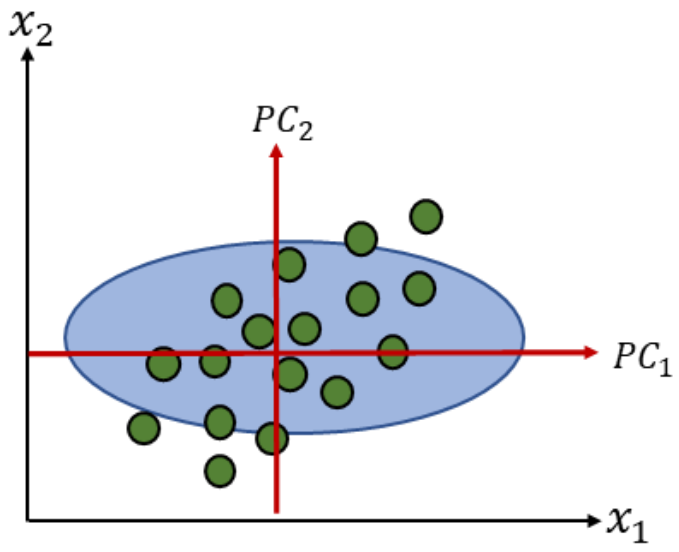
- Summarizing: Construct frequency tables or summary statistics
- Visualizing: Construct appropriate graphs to examine distribution of data
  - Box plots; Histograms; Bar charts
- Correlation: Examine variable relationships
  - Correlation; Scatter plots
- Perform dimension reduction or cluster analysis

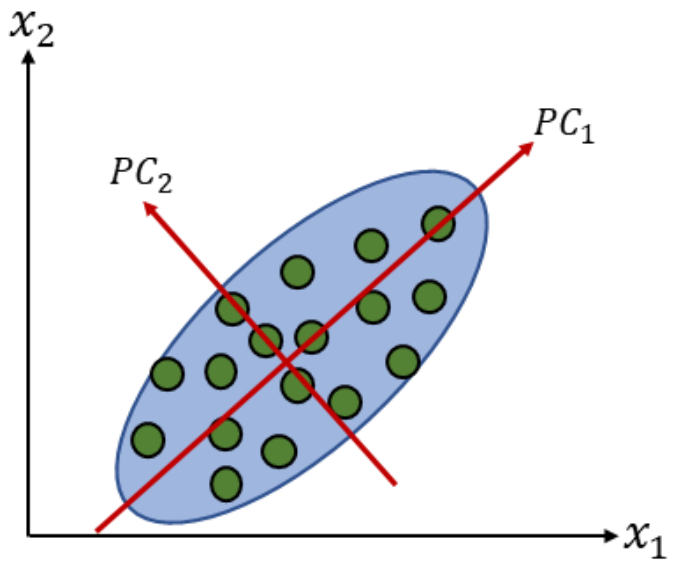




Dimension Reduction

- Background:
  - Raw data are often sparse as a consequence of the curse of dimensionality
  - Enormous amount of data is generated daily, e.g. Facebook, Amazon, smartphone, etc.
  - Visualization of data and drawing inferences becomes more and more challenging
- Dimension Reduction aims at transforming data from a high-dimensional space into a low-dimensional space so that the low-dimensional space can retain "meaningful" properties of the original data.
- Types of Dimension Reduction Techniques:
  - Linear technique: Principal Components Analysis (PCA)
  - Nonlinear technique: t-Distributed Stochastic Neighbor Embedding (t-SNE)
- What is Principal Components Analysis (PCA)?
  - Principal component analysis (PCA) is the process of computing the principal components (which are the linear combinations of observed variables) and using them to perform a change of basis on the data.
  - Simply speaking, in the 2-dimensional input space, the principal components (PC_1 and PC_2) are obtained by rotation of axis and shifting of the origin.



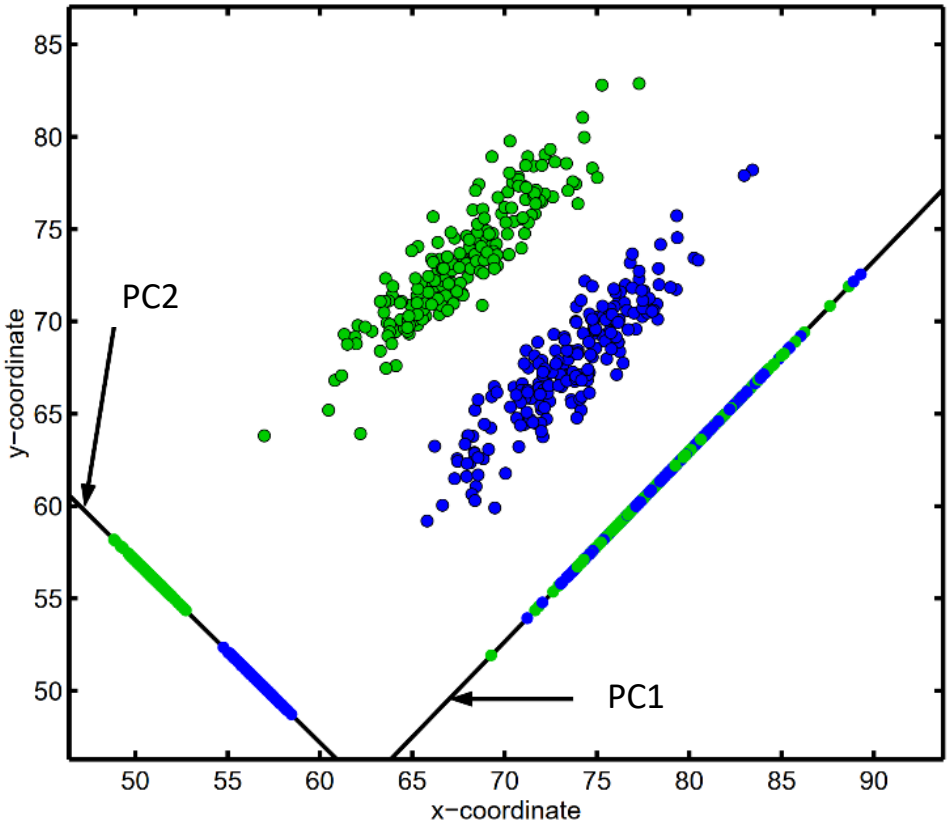


Principal Component Analysis (PCA)

STEM Activity: Concept of Projection

Go to https://in-the-sky.org/ngc3d.php, disable Auto-rotate and Show labels and select Zoom to show Local Milky Way. Now let's rotate the picture of the galaxy to find the layout of the stars.

- Aim to produce a small number of linear combinations $\{Z_1, \dots, Z_p\}$ (called principal components (PC)) of the original inputs $\{X_1, \dots, X_p\}$
- Each PC $Z = a_1 X_1 + \cdots + a_p X_p = \boldsymbol{a}'\boldsymbol{X}$ is found to maximize the variance of Z and is uncorrelated with all the previous PCs.
- Note that PCA is a linear method. A nonlinear method is t-Distributed Stochastic Neighbor Embedding (t-SNE)



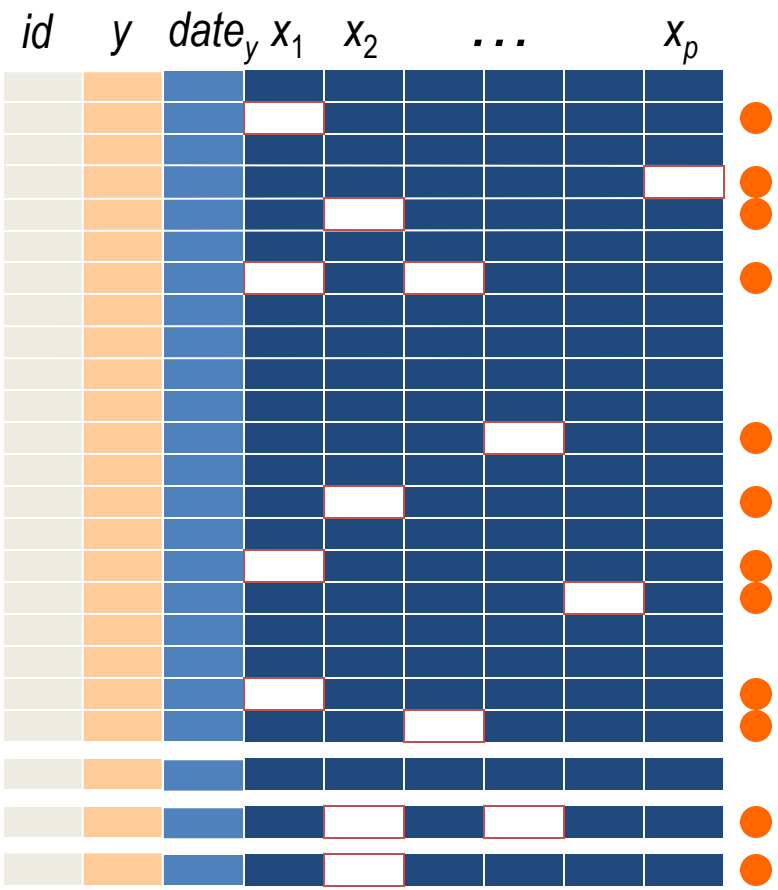
Missing Values

Possible Reasons:

- Hardware failure
- Nondisclosure by respondents

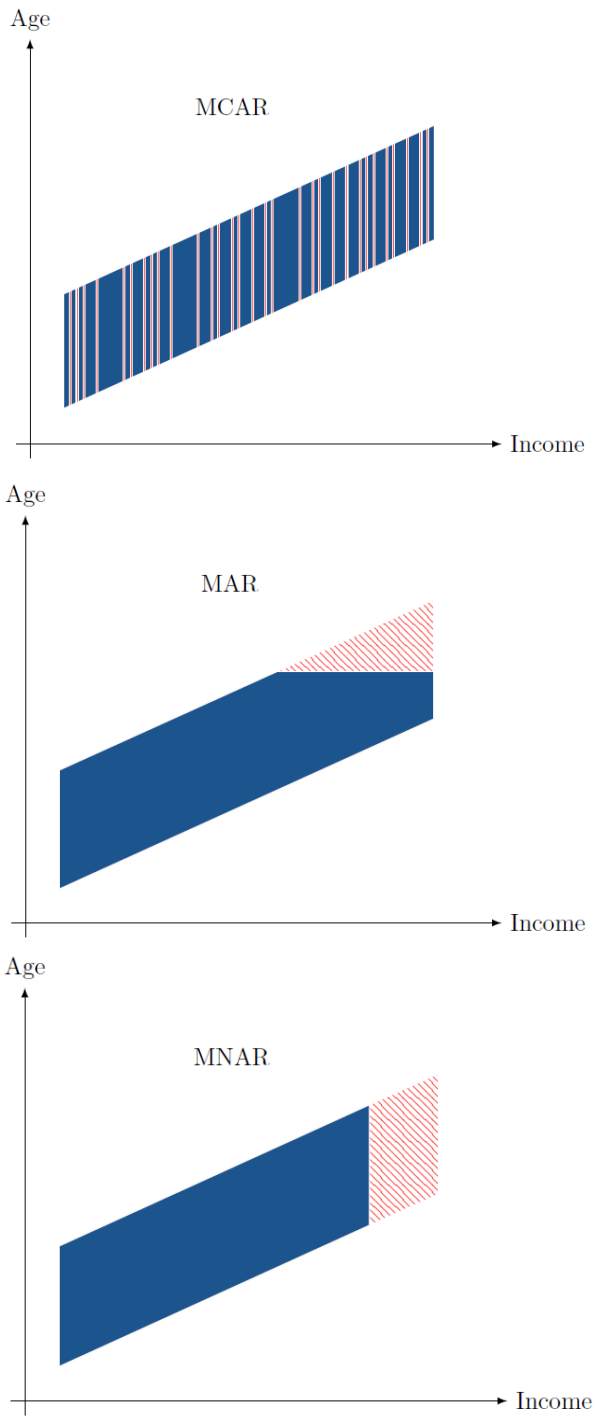
How to Handle Missing Data?

1. Go find the missing data
   - May be Costly + Time Consuming + Infeasible?
2. Ignore the whole record (listwise deletion)
   - Simplest solution
   - Drawbacks: may ignore useful or important information from the non-missing variables and may bias the sample
   - May be fine for few missing values
   - Other variant: pairwise deletion
3. Treat missing values as 'Unknown'
4. Impute (fill in) the missing values
5. Use models which support missing values. Eg. 1-NN, simple tree

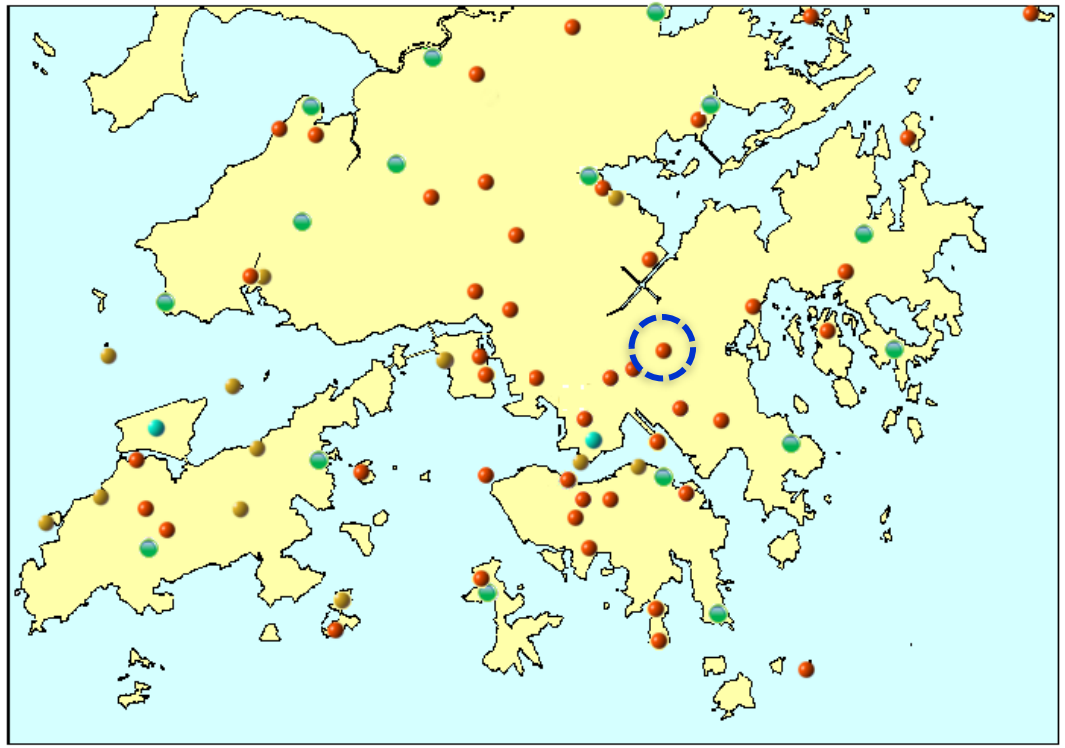


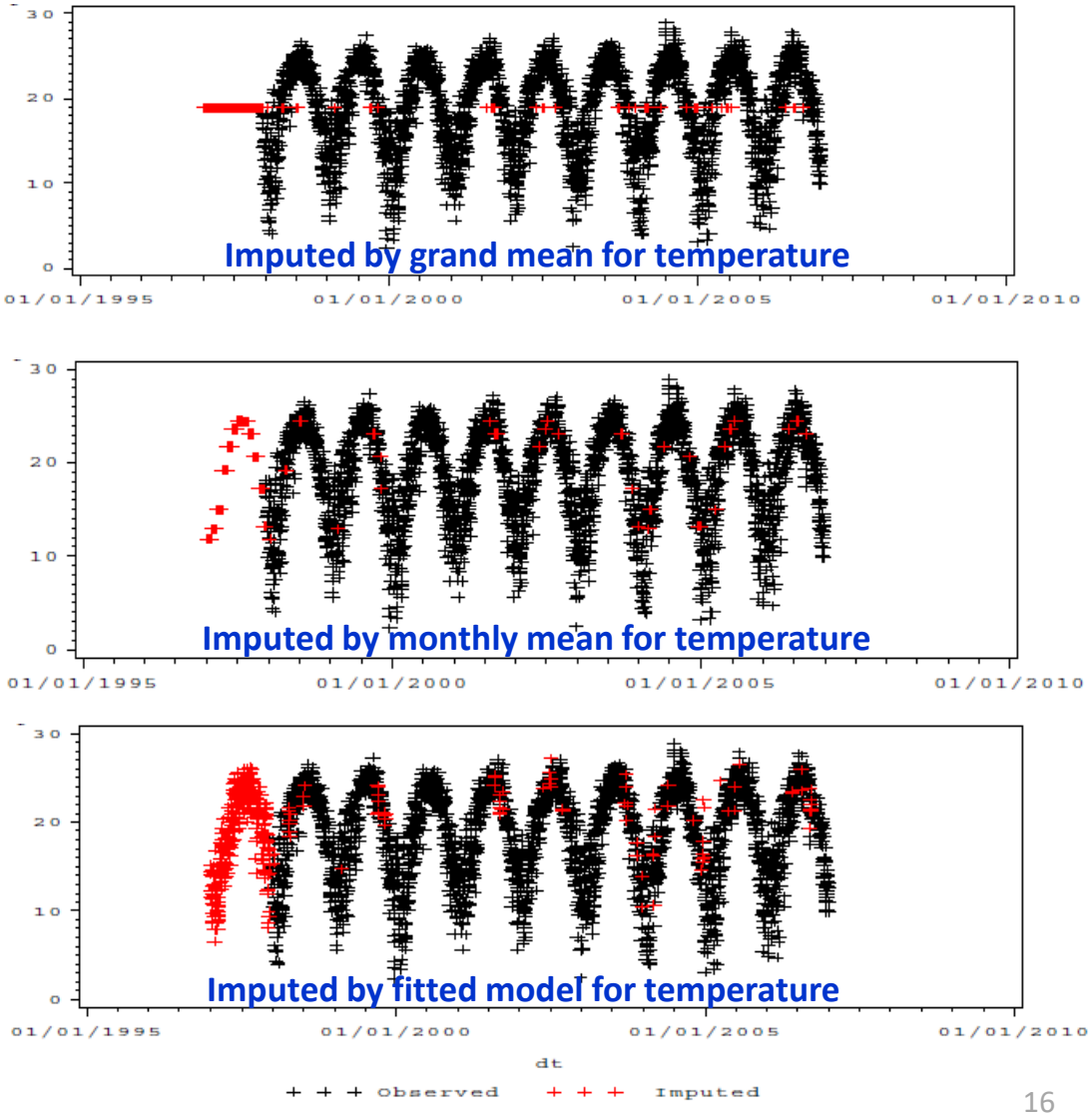
Missing Values

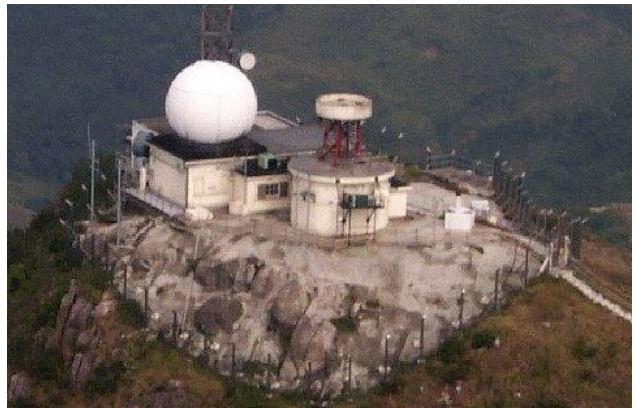
- ✓ Data $Y = (Y_{obs}, Y_{mis})$
- ✓ R indicates whether a data point is missing(=0) or not(=1)
- ✓ Rubin (1976) classified missing data into three categories:
  - ✓ A dataset is said to be Missing Completely at Random (MCAR) if
  - $P(R=0 \mid Y_{obs}, Y_{mis}, \theta) = P(R=0 \mid \theta)$
    - Listwise deletion, pairwise deletion and mean imputation
  - ✓ A dataset is said to be Missing at Random (MAR) if
  - $P(R=0 \mid Y_{obs}, Y_{mis}, \theta) = P(R=0 \mid Y_{obs}, \theta)$
    - Model-based imputation (missing indicator)
  - ✓ Otherwise, the dataset is said to be Missing Not at Random (MNAR)



Tate's Cairn weather station 大老山氣象站







Approaches of Handling Missing Data

- ✓ Data Deletion
  - ✓ In the case of MAR, we can use list-wise or pair-wise deletion of the missing observations. However, this option may not be feasible if huge portion of missing.
- ✓ Data Imputation
  - ✓ Mean or Median Imputation
    - ✓ Use the mean or median of the non-missing observations.
    - ✓ It can be used when the number of missing observations is small.
    - ✓ When the number of missing observations is large, using mean or median can result in loss of variation in data.
  - ✓ Model-based Imputation
    - ✓ K-NN Imputation
    - ✓ Multivariate Imputation by Chained Equations (MICE)
  - ✓ Many others…

Six blind men and an elephant

An Indian story: a group of blind men touch an elephant to learn what it is like.

- The blind man who feels a leg says the elephant is like a pillar;
- the one who feels the tail says the elephant is like a rope;
- the one who feels the trunk says the elephant is like a tree branch;
- the one who feels the ear says the elephant is like a hand fan;
- the one who feels the belly says the elephant is like a wall; and
- the one who feels the tusk says the elephant is like a solid pipe.

- **The king explains to them:**
  - **All of you are right. The reason every one of you is telling it differently is because each one of you touched the different part of the elephant. So, actually the elephant has all the features you mentioned.**

Source: http://en.wikipedia.org/wiki/Blind_men_and_an_elephant

Feature Transformation and Selection

- Feature transformation:
  - Transformation of data to improve the accuracy of the algorithm
  - Eg. Polynomial/Interaction, Min-max normalization/Standardization, Binning, Imputation
- Feature selection:
  - Selecting relevant features and removing unnecessary features, E.g.:
    - Select features strongly correlated with the target
    - Selection by modeling: LASSO, Random forest
    - Dimensional reduction by Principal Component Analysis (PCA)
    - Many others…

Why Transformation?

- Variables tend to have ranges different from each other
  - In a credit card customer database,
    - No. of credit cards: [1, 10]
    - Amount of credit limit: [5,000, 100,000]
  - Some data mining techniques such as cluster analysis and dimension reduction methods adversely affected by differences in variables' ranges
  - Solution: to normalize or standardize the variable
- To create new variables from existing ones. Eg. Aggregation
- To group rare levels for a discrete variable
- To correct non-normality in a variable
- To discretize continuous variables (say, age to age group) (discretization or binning)

Two Normalization Methods

- Min-Max Normalization (Range Normalization)
  - To scale the values of a variable to fall within a small, specified range (say [0,1])

$$X^* = \frac{X - \min(X)}{\text{range}(X)} = \frac{X - \min(X)}{\max(X) - \min(X)}$$

- Z-Score Standardization
  - To scale the values of a variable to have mean 0 and variance 1

$$X^* = \frac{X - \text{mean}(X)}{\text{SD}(X)}$$

- Example: {9, 14, 25}
  - Min-max: {0, 0.3125, 1}
  - Z-score: mean = 16, SD = 8.185, {-0.855,-0.244,1.099}
- Mean and variance are sensitive to outliers. In this case, use robust estimates: median and interquartile range.

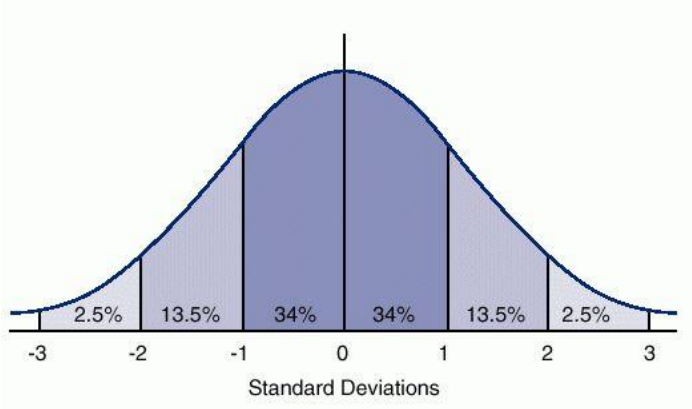
Binning (Discretization)

- Transform a continuous variable to an ordinal variable
- K-bins discretization (unsupervised)
  - Equal-width binning (uniform)
    - Divide data values into k equally spaced intervals
    - Scott's formula for optimal no. of bins, $k = (max - min)/w$ where $w = 3.5sN^{-1/3}$, s = sample SD, N = data size
  - Equal-frequency binning (quantile)
    - Divide data values into k bins with approximately the same frequency in each bin
  - Clustering (kmeans)
    - Apply k-means clustering to group the data values.
- Normally inferior to supervised methods in classification tasks.

Class Activity 2:
Data Handling Issues

Remarks

- Missing data
  - Missing data may be useful for prediction.
  - Missing is missing! No perfect solution but we hope to minimize the impact of missing data to the study
- Outliers
  - No clear definition of outliers!
  - Outliers could be detected because of using a wrong model.
  - Outliers could be good ?! e.g. Credit-card fraud detection relies a lot on outliers (abnormal spending patterns) detection to discover possible cases of fraud.
- Feature (Variable) selection
  - The methods considered here assume no missing values. In recent years, methods have been proposed to allow variable selection with missing data.



Remarks

- Ethical issues
  - Data privacy issues (私隱問題)
  - Discrimination bias (歧視偏見)

Gender bias of word embedding:
woman + king - man ≈ queen can do a magic but do you like the following?
woman + doctor - man ≈ nurse

Source: https://towardsdatascience.com/gender-bias-word-embeddings-76d9806a0e17

TIME STAMP: 2030.07.14

"It's time to wake up, Sue," says FIA (acronym for Financial Intelligent Agent). "You weren't sleeping well. Your happiness quotient is lower today. Would you like to sleep an extra hour and we'll talk again at 9 a.m.?"

Sue says yes and sleeps until her scheduled meeting. By the time 9 a.m. arrives, she is dressed, drinking a cappuccino, which her smart kitchen began brewing for her the moment she exited the shower. Now ready to face the day, Sue listens to FIA through a mind interface.

"While you were sleeping, I performed a self-upgrade, enhancing my participation in a 360-degree holistic view of your life. This helps me improve your well-being and your financial status—a top priority for you."

Sue nods her head as she sips her drink and stares out the window.

"I am making continual micro-adjustments to your investment portfolio based on dynamic trends being monitored on a continuous 24/7 basis," FIA continues. "I am redistributing your savings through micro-auctioning to the Top Ten-ranked services for best spot interest rates."

Sue suddenly realizes she forgot to schedule a flight to her nephew's birthday party next month and thinks her concern.

"Also, no need to fret over missing little Tommy's get-together. I examined your calendar and through UIA (United Intelligent Agent), I booked flights for August 20 at the lowest rates."

"Whew. Thank you, FIA!"

"Also, I am sensing a 90 percent probability for a high flu outbreak in Amsterdam when you arrive. Using IIA (Insurance Intelligent Agent), I was able to negotiate spot health insurance for the 23.3 hours of risk."

"That's smart. No use in paying for the whole weekend if I don't need it."

"Next, I booked you an autonomous rental car with the best safety record, thus the lowest insurance. You can call him SAM. Your hotel is handled the same way through their intelligent agents (IAs)."

"Perfect."

"And just so you know, I will handle via IAs all of the fund transfers and payments with the best-negotiated lowest fees. From my analysis of your thought patterns, I know you worry about this. We can get all this information in one place, updating continuously, with the best deals and rates, lowest fees, no minimums, and no overhead infrastructure since it is all virtual."

Class Activity 3:

Lab 2: Data Preprocessing

M A U Q R I Z
O T H A N K A
Y O U N D S T
X P V E R Y A
O M U C H N Z
A N L T Y O U
R I G W P I T
E T H E K O N
F O M B E S T